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Research Article

Asymmetrical interlingual influence in the production of Spanish and English laterals as a result of competing activation in bilingual language processing

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ABSTRACT

This study examines the phonetic and phonological knowledge of Spanish and English /l/ by early and late Spanish-English bilinguals along a continuum of language dominance. Forty early Spanish-English bilinguals, divided into groups as a function of their immigrant generation (G1.5, G2, G3), and twenty L2 Spanish learners produced word-initial and word-final laterals in three separate sessions: monolingual Spanish session, monolingual English session, and bilingual Spanish/English session. Results indicate that all participants acquired the phonetic and allophonic characteristics of the lateral variants in each language, and that language dominance strongly predicts these bilinguals' acoustic realization of Spanish and English laterals. The acoustic analyses reveal phonetic convergence as a result of language mode, but it is especially the laterals in the non-dominant language that are altered in bilingual mode. This study provides evidence of language dominance effects on the acoustic realization of Spanish and English laterals, and demonstrates the impact of language mode on the phonetic abilities of early and late bilinguals. It is proposed that a model of bilingual processing based on the principles of episodic frameworks, in which exemplars in a bilingual lexicon develop connections across languages and lexical processing activates the words of both languages non-selectively, can explain these findings.

1. Introduction

Bilingual individuals must have the ability to produce language-specific acoustic targets accurately and consistently, and the challenges in reducing cross-linguistic phonological influence will largely depend on the L1–L2 acoustic relationship of the speech sound that is going to be articulated in the target language. For instance, a Spanish-English bilingual individual will have to realize (i) phonological contrasts that exist in only one of their languages (e.g., Spanish trill or the English /ɪ/-/iː/ contrast), (ii) language-specific contextual/distributional phonological processes (e.g., Spanish spirantization), and (iii) phonological categories that exist in both languages but are slightly different, either at the acoustic level or in terms of the temporal coordination of articulatory gestures (e.g., VOT in word-initial voiceless stops). In addition to articulating these language-specific speech sounds when producing each language, the bilingual individual will necessarily have to produce them while inhibiting or deactivating the other language (Green, 1998).

The most influential theoretical models in the area of L2 phonological acquisition (Perceptual Assimilation Model, PAM, Best, 1995; Perceptual Assimilation Model of Second Language Speech Learning, PAM-L2, Best & Tyler, 2007; the Speech Learning Model, SLM, Flege, 1995; and the Second Language Perception Model, L2LP, Escudero, 2005) assume that the learnability of new speech sounds in the L2 is perceptual in nature and depends on the perceived phonetic distance between the sounds in the L2 and the most similar sounds in the L1 phonetic inventory. For instance, the SLM (Flege, 1995) emphasizes the close link between speech perception and production by assuming that accurate perception of the phonetic differences between two L2 phones can eventually lead to the correct production of these differences. In particular, the SLM presents two possibilities: (1) if the learner identifies the foreign stimulus as a similar phone, then an L1 category will be used to represent that phone in production, or (2) if the L2 speaker considers that the foreign speech sound is a new phone, then the L2 learner will eventually create a new category different from the categories already existing in the L1. In the first case, the model hypothesizes that L2 be assimilated to an equivalent speech sound preventing learners from establishing a separate phonetic category. In the second case, the hypothesis is that a new category will be formed because it cannot assimilate to an L1 category. Finally, the SLM postulates that early learners, as opposed to adult L2 learners, are capable of establishing additional phonetic categories for similar L2 speech sounds, as early learners' L1 phonetic categories are malleable while late learners' L1 categories are already fully developed. Thus, the SLM's interaction hypothesis predicts that the speech sounds of the L1 and L2 are less likely to interact in early bilinguals than in late bilinguals (Flege, 1992).

While most studies have shown that early learners outperform late learners in various production and perception tasks (Baker & Trofimovich, 2005; Darcy & Krüger, 2012; Flege, Yeni-Komshian, & Liu, 1999), the source of age effects still remains controversial (Flege & MacKay, 2004). Several studies on bilingual speech production have observed that early bilinguals produce and perceive L1 and L2 speech sounds free of interlingual interference (Flege, MacKay, & Meador, 1999; Guion, Harada, & Clark, 2004; Mack, 1989; Piske, Flege, MacKay, & Meador, 2002), suggesting that early exposure helps to develop and maintain independent or separate phonetic systems. However, a number of studies suggest that these bilinguals do not necessarily realize target-like speech sounds in the L2; rather, the bilinguals' combined or interrelated systems influence each other at a fine-grained acoustic level (Flege, Munro, & MacKay, 1995; Flege et al., 1999; Fowler, Sramko, Ostry, Rowland, & Hallé, 2008; Piske et al., 2002), and early and extensive exposure to a second language may not be sufficient to attain target-like phonetic abilities in the language (Bosch, Costa, & Sebastián-Gallés, 2000; Pallier, Bosch, & Sebastián-Gallés, 1997; Sebastián-Gallés, Echeverría, & Bosch, 2005; Sebastián-Gallés & Soto-Faraco, 1999). In order to better understand the phonological/phonetic systems of early bilinguals, researchers necessarily have to consider many variables in addition to the age of acquisition, such as language proficiency, language dominance, language use and other non-linguistic variables that are particular to the experiences of the bilingual community under investigation (Amengual, 2012, 2016a, 2016b, 2016c; Amengual & Chamorro, 2015; Flege, Frieda, & Nozawa, 1997; Flege, Schirru, & MacKay, 2003; Guion, 2003; Simonet, 2014, 2015).

The present study examines the acoustic realization of a phonemic category that exists in both languages of a Spanish-English bilingual but that differs in its phonetic realization and allophonic patterning in the two languages. Specifically, this study analyzes the acoustics of the Spanish and English voiced lateral approximant (i.e., /l/) as produced by four groups of Spanish-English bilinguals with markedly different linguistic backgrounds: three groups of early Spanish-English bilinguals divided as a function of their immigrant generation (G1.5, G2, and G3), and a group of late-onset L2 Spanish learners (L1 English) who learned Spanish in a classroom setting. This experiment investigates factors not previously addressed simultaneously on heritage speakers and L2 Spanish context-specific allophonic patterning in Spanish and English, which in these bilingual groups, involves the inhibition of an allophonic distinction in English, as opposed to the acquisition of a new phonemic category.

2. Background

2.1. Heritage language phonetic and phonological acquisition

Benmamoun, Montrul, and Polinsky (2013) characterize *heritage speakers* as asymmetrical bilinguals who learned the heritage language as an L1 in childhood, but who, as adults, are dominant in a different language. In the context of the United States, the term heritage speaker has been used to refer to an individual who is raised in a home where a non-English language is spoken, who speaks or merely understands the heritage language, and who is to some degree bilingual in English and the heritage language (Valdés, 2000, 2005). In other words, heritage speakers are early bilinguals who have been exposed to the minority (heritage) language and the majority language early in life, either by growing up speaking both languages since birth or having been brought up in a monolingual setting in early childhood and becoming bilingual after starting school in the majority language at around ages 5 or 6. The linguistic abilities of heritage speakers have been compared in recent years to both monolingual speakers and to second language (L2) learners showing that heritage language acquisition typically results in a non-target-like competence and use of the language, with transfer from the dominant language, a better ability with receptive than productive language, and linguistic gaps that resemble the patterns attested in second language acquisition (Montrul, 2011; Montrul, Bhatt, Bhatia, & Girju, 2012; O'Grady, Kwak, Lee, & Lee, 2011; Polinsky, 2006; Rothman, 2007).

Although in the last few decades there has been an increase in the number of studies investigating the linguistic abilities of Spanish heritage speakers, heritage Spanish phonetics and phonology has received considerably less attention than other linguistic subfields, especially in comparison to research focusing on heritage language morphosyntactic knowledge or studies aimed at devising pedagogy tailored to meet heritage language learners' linguistic needs (Potowski, 2013; Rao & Ronquest, 2015). The general assumption has been that Spanish heritage speakers have a benefit in pronunciation as a result of early exposure to the minority language (Au, Oh, Knightly, Jun, & Romo, 2008; Chang, Haynes, Rhodes, & Yao, 2008; Knightly, Jun, Oh, & Au, 2003), but recent studies have also shown differences between the pronunciation of heritage speakers and monolingual speakers of the minority language (Au, Knightly, Jun, & Oh, 2002; Bullock, 2009; Chang, Haynes, Yao, & Rhodes, 2009; Ronquest, 2012). Since there are still relatively few studies in the phonological/phonetic domain that have empirically investigated the Spanish pronunciation of these early bilinguals (Amengual, 2012, 2016c, 2017; Henriksen, 2015; Rao, 2014, 2015; Ronquest, 2012), at this point we cannot claim that we community characterized by a shift from Spanish to English across three generations, in a situation in which Spanish is socially and linguistically subordinate to English. Even though Silva-Corvalán (1994) reports that third-generation Spanish speakers in Los Angeles were found to have considerably reduced Spanish verbal systems, she argues that the morphosyntactic component seems to be quite resistant to change (Silva-Corvalán, 1998), and contact-induced change mostly affects the lexicon and the discourse-pragmatics level. The current study examines the phonetic production of a speech sound that exists in both Spanish and English but is slightly different in both languages of the bilingual individual as a function of the speaker's immigration generation. Due to a well-documented process of language shift from Spanish to English (Bills, Hernández-Chávez, & Hudson, 1995; Martínez, 2009; Potowski, 2004), immigrant generation is expected to strongly correlate with language dominance.

2.2. Language dominance

Bilingual individuals, especially early bilinguals, very frequently demonstrate a high level of proficiency in both of their languages. However, the perfectly balanced bilingual probably does not exist: bilinguals have a dominant or stronger language (Cutler, Mehler, Norris, & Seguí, 1986; Flege, MacKay, & Piske, 2002; Silva-Corvalán & Treffers-Daller, 2016). Language dominance refers to "the observed asymmetries of skill in, or use of, one language over the other" (Birdsong, 2014, p. 374) and it is a variable that has been found to have an impact on code-switching patterns (Basnight-Brown & Altarriba, 2007), govern bilingual lexical memory representation (Heredia, 1997), and determine the language of mental calculations (Tamamaki, 1993). Of specific interest to this study, language dominance has also been shown to account for cross-linguistic phonological influence in bilingual speech production (Amengual, 2016a, 2016b, 2016c; Bullock, Toribio, González, & Dalola, 2006; Simonet, 2011, 2014), perception (Antoniou, Tyler, & Best, 2012; Bosch et al., 2000; Pallier et al., 1997; Sebastián-Gallés & Soto-Faraco, 1999), and processing (Amengual, 2016d; Pallier, Colomé, & Sebastián-Gallés, 2001).

The operationalization of language dominance has been an especially challenging task due to the fact that it covers many dimensions of language use and experience, such as proficiency, fluency, ease of processing, frequency of use, and cultural identification. A considerable number of procedures have been used to assess bilingual language dominance, including global ratings of fluency (Tamamaki, 1993), the subjective judgment of a bilingual experimenter (Talamas, Kroll, & Dufour, 1999), scores obtained on vocabulary tests in the L1 and L2 (Cromdal, 1999), language history (Hazan & Boulakia, 1993), and language used at home (Sebastián-Gallés et al., 2005). While it must be acknowledged that there is currently no uniform, widely accepted method for assessing bilingual dominance (see Dunn & Tree, 2009; Li, Zhang, Tsai, & Puls, 2014; Marian, Blumenfeld, & Kaushanskaya, 2007; Runnqvist, Montoya, & Cera, 2012; Lim, Liow, Lincoln, Chan, & Onslow, 2008). As opposed to objective tests of language ability, self-assessments of language abilities and experience succeed in accounting for non-linguistic factors, such as language attitudes, which are an important aspect of dominance (Pavlenko, 2004).

Language dominance in the present study is calculated from participant responses to the Bilingual Language Profile questionnaire (BLP; Birdsong, Gertken, & Amengual, 2012), an instrument for assessing language dominance through self-reports that produces a continuous dominance score and a general bilingual profile taking into account multiple dimensions: age of acquisition of the L1 and L2, frequency and contexts of use, competence in different skills, and attitudes towards each language. This assessment tool has been used in recent studies to classify bilingual participants as a function of their language dominance in a variety of language pairs, such as Spanish-Catalan (Amengual, 2016a, 2016b; Nadeu & Renwick, 2016; Simonet, 2014), French-English (Birdsong & Gertken, 2013), Spanish-K'ichee' (Baird, 2015), Spanish-Yucatec Maya (Uth, 2016), Spanish-Afrikaans (Coetzee, García-Amaya, Henriksen, & Wissing, 2015), Spanish-Galician (Amengual & Chamorro, 2015), Spanish-English (Amengual, 2016c, 2017; Casillas, 2015; Casillas & Simonet, 2016), Mandarin-English (Dmitrieva et al., 2015), and Spanish-Welsh (Bell, 2017).

2.3. Language mode

Language mode is the state of activation of the bilingual's languages and language processing mechanisms at a given point in time (Grosjean, 2008). The language mode framework (Grosjean, 1998, 2001, 2008) posits that, depending on the communicative situation, just one of a bilingual's languages may be activated (monolingual mode), or both languages may be activated (bilingual mode). This activation is determined by psychosocial and linguistic factors as a consequence of the communicative context: a monolingual mode will arise when the interlocutor or the situation requires that only one language be used to the exclusion of the other, whereas a bilingual mode will arise if the interaction occurs between two bilingual individuals who share the same language and who feel comfortable mixing languages, or when a bilingual is listening to a conversation which contains elements of the other language. As stated in Grosjean (2001, p. 2), "In the monolingual speech mode, the bilingual deactivates one language (but never totally) and in the bilingual mode, the bilingual speaker chooses a base language, activates the other language and calls on it from time to time in the form of code-switches and borrowings".

There is still a relatively small body of work that examines language mode effects on the phonetic interactions between the two languages of the bilingual individual. These studies have investigated the effects of linguistic code-switching on speech production, but the evidence, so far, for the effects of a bilingual's language modes on their phonetic behavior is and insertional code-switches. Bullock et al. (2006) found that Spanish-English bilinguals produced distinct categories for English and Spanish stops, however, English stops moved towards the Spanish VOT range in the code-switching context. Evidence for a language mode effect at the phonetic level was found in Antoniou, Best, Tyler, and Kroos (2010, 2011) with Greek-English bilinguals. The results indicated that monolingual mode elicited VOTs that were in the monolingual range for each language but the code-switching context revealed values in the L2 that were affected by the L1. Similar language mode effects were found in Olson (2012), where Spanish-English bilinguals hyperarticulated their productions in code-switched sentences, with higher pitch and increased duration in insertional code-switched words. Finally, Simonet (2014) investigated the production and perception of the Catalan back mid-vowel contrast by Spanish-dominant and Catalan-dominant Spanish-Catalan bilinguals in two different experimental sessions (unilingual vs. bilingual session). The results demonstrated that the Spanish-Catalan bilingual participants, regardless of their language dominance patterns, demonstrated increased cross-linguistic convergence in the bilingual session, compared to the unilingual session. The present study adds to the body of work that manipulates the experimental context in order to increase the effects of transient interlingual interference by examining the effects of language mode on the acoustic realization of Spanish and English laterals in three different sessions (monolingual English, monolingual Spanish, and bilingual English/Spanish session).

2.4. The phonetic variable: Laterals in Spanish and English

The lateral approximant /l/ is a phoneme that is shared between Spanish and English but the lateral in each language differs in terms of its acoustic–phonetic properties and its allophonic distribution. Spanish /l/ is typically described as being "clear" [l], in that it is perceived as more consonantal in quality. The /l/ in American English is characterized as being relatively "dark" [ɫ], in that it is perceived as more vowel-like in quality (Barlow, 2014; Huffman, 1997; Olive, Greenwood, & Coleman, 1993). American English /l/ is produced through two co-occurring gestures: a primary articulation including contact between the tongue tip and the alveolar ridge and a secondary articulation of the tongue dorsum arching upward at the velum. It is this latter vocalic tongue-dorsum gesture that results in what is known as velarization or darkness (Sproat & Fujimura, 1993). These laterals manifest acoustically via differing resonant frequencies, and specifically clear and dark /l/ diverge in the second formant (F2). Specifically, active dorsal backing and lowering in dark laterals triggers a lowering of the F2 and a raising of the F1 as compared with clear laterals (Recasens, 2004). Therefore, a clear /l/ has a high F2 value and a large difference between F2 and F1 whereas dark /l/ is associated with lower F2 values and a smaller F2–F1 difference (Carter, 2003; Proctor, 2009; Recasens, 2004, 2012; Simonet, 2010, 2015; Turton, 2017; Yuan & Liberman, 2011).

Additionally, Spanish and English laterals differ in terms of dorsal backing and lowering, and lower F2 and higher F1 values, being produced darker than laterals in syllable onset position (Sproat & Fujimura, 1993). For instance, [ɫ] occurs in coda positions (e.g., seal [si:ɫ]), and [l], surfaces in the syllable onset (e.g., leave [li:v]). In other words, laterals are typically "darker" in syllable rhyme and "clearer" in syllable onset. Some varieties of American English are also reported to show vocalization of dark /l/ in syllable rhymes (Ash, 1982; Pederson, 2001); however, it is not as widespread as in the varieties of English spoken in South-Eastern Britain (Trudgill, 1986; Wells, 1982). In contrast to English, the Spanish /l/ phoneme does not vary by syllabic context (e.g., lago [laɣo] 'lake' and mal [mal] 'bad'). Therefore, "clear" /l/ in Spanish is assumed to be relatively consistent across syllabic contexts (pre-vocalic, post-vocalic). Another factor that has been shown to affect the degree of velarization of the alveolar laterals is the vowel context. Data from Eastern Catalan indicate that dark laterals may become less velarized, and as a result be produced with less darkness, in the context of a high fronted vowel such as /i/, while clear laterals may become less clear in the context of /u/ (Recasens, 2004; Recasens & Espinosa, 2005; Simonet, 2010, 2015).

Previous studies on the production of the lateral approximant in Spanish-English bilinguals have revealed cross-linguistic influence in their speech as a result of factors such as age of acquisition, language proficiency, and elicitation task. Barlow, Branson, and Nip (2013) analyzed pre-vocalic and post-vocalic laterals produced by Spanish-English bilingual, Spanish monolingual, and English monolingual children in order to determine if Spanish-English bilingual children produce language-specific lateral approximants or if they produce a lateral that is intermediate to that of Spanish and English monolinguals. The results indicated that Spanish-English bilingual children formed separate phonetic categories (i.e., distinct phonetic categories) for postvocalic /l/ in Spanish and English but that they produced a merged phonetic category for /l/ in prevocalic position. In a production study with Spanish-English bilingual adults, Barlow (2014) demonstrated that age of acquisition impacts bilinguals' production of Spanish and English laterals. In this study, the results indicated that even though the English laterals were affected due to cross-linguistic influence from Spanish regardless of the age at which English was acquired, this effect was significantly stronger for those who learned English at a later age. Additionally, the influence of English on Spanish /l/ production was only apparent for late bilinguals but not early bilinguals.

Solon (2016) investigated the acoustic realization of /l/ by five groups of L2 Spanish learners divided as a function of their Spanish language level (first year, second year, third year, fourth year, and graduate students). Their productions were compared to those of a native speaker control group. As expected, the results showed that the productions in Spanish were gradually lighter (i.e., more Spanish-like) as the Spanish level increased and that all speaker groups produced more velarized laterals when following a back vowel than when the lateral followed a front vowel. It was also speakers. Therefore, language proficiency was a strong predictor of the ability to inhibit contextual patterns in English and produce Spanish laterals with no differentiation based on syllable position. The factors of elicitation task and preceding vowel interacted with Spanish language level, with the impact of experimental task (i.e., picture description, sentence reading, and word-list reading) especially influencing lower level learner groups.

3. The present study

The present study examines the acoustic realization of voiced lateral approximants in the Spanish and English of heritage Spanish speakers and L2 Spanish learners, but goes beyond Barlow et al. (2013), Barlow (2014), and Solon (2016) in three ways: (i) as in Barlow (2014), it examines the production of early and late Spanish-English bilinguals, but takes into consideration the variable of immigrant generation (G1.5, G2, G3) in the early Spanish-English bilingual group to provide a more fine-grained analysis of this heterogeneous bilingual population, (ii) it complements the data in Barlow (2014) and Solon (2016) by investigating the effects of language dominance in the English and Spanish production of word-initial and word-final laterals by each of these four speaker groups, and (iii) it adds the variable of language mode to the analysis of Spanish and English /l/, a factor that has not been examined in previous studies on the phonetic and allophonic acquisition of these lateral variants. The main research questions that are explored in this production experiment are the following:

1. Do early and late Spanish-English bilinguals produce acoustically similar Spanish and English lateral approximants or do they maintain separate acoustic realizations for each language?
2. In addition to acquiring the acoustic properties of the laterals in Spanish and English, do these bilinguals successfully acquire the language-specific contextual (allophonic) patterns of use of these variants?
3. Is there evidence of phonetic convergence of the laterals in each language as a result of eliciting a bilingual language mode? Does language dominance modulate the acoustic realization of these laterals?

Based on the results from previous studies on the effects of age of acquisition in bilingual speech production, it is hypothesized that speakers who acquired Spanish earlier will produce lighter /l/'s, that is, with smaller F2–F1 values in their Spanish and English productions, than native English-speakers who acquired Spanish later in life. Therefore, the L2 Spanish learners are expected to produce less target-like Spanish laterals (with higher F2–F1 values) and produce darker laterals in English in comparison to the early Spanish-English bilingual groups (G1.5, G2, G3). In terms of the SLM (Flege, 1995), some English-dominant speakers who learned Spanish later in life, particularly from the L2 Spanish learner group, might have difficulties establishing a new L2 category for these sounds giving rise to a merged L1 and L2 phonetic category ("equivalence classification"). In line with the predictions from

In addition to acquiring a new phonetic category, these individuals also have to learn the different contextual patterning of Spanish and English /l/. In English, different lateral variants (light, dark) are realized in syllable-initial and syllable-final positions, but this variation by syllable position does not occur in Spanish. Results from previous studies lead us to competing hypotheses. On the one hand, Barlow (2014) finds evidence of interlingual influence from Spanish to English in the productions of early bilinguals. The late bilingual group show this phonetic influence in the same direction but to a greater degree; however, these individuals also show interlingual influence from English to Spanish. Crucially, for these bilinguals the English allophonic rule also applies to their Spanish productions. On the other hand, Solon (2016), as in Barlow (2014), finds that laterals produced in coda position in Spanish by first-, second-, and third-year learner levels are significantly darker than laterals produced in onset position, presenting evidence that this context-specific allophonic patterning in English had been transferred to their Spanish. However, no significant distinctions between onset and coda laterals in Spanish were found for the fourth-year level or the graduate-level learners. This indicates that after one has achieved a sufficiently high Spanish language proficiency, it is much more likely that one can inhibit the allophonic contextual patterns from English in their Spanish. These studies on the production of laterals by Spanish-English bilinguals have examined age of acquisition and language proficiency, but not language dominance. The present study investigates the effects of language dominance on the production of context-specific allophonic patterning in Spanish and English.

The novelty of the present study is that it examines how language dominance interacts with the linguistic background of Spanish heritage speakers of different immigrant generations (G1.5, G2, G3) and L2 learners producing laterals in both of their languages. Based on the results from previous studies demonstrating an effect of language dominance on bilingual speech production (Amengual, 2016a, 2016b, 2016c; Amengual & Chamorro, 2015; Bullock et al., 2006; Simonet, 2011, 2014), it is predicted that the degree of language dominance will play a substantial role on the degree of velarization of the laterals in both Spanish and English, operationalized as the F2–F1 difference. Spanish-English bilinguals who are more dominant in English are expected to produce a darker lateral (smaller F2–F1 difference) in syllable-final position than syllable-initial position in English, and if they do not inhibit this distinction they will transfer the English contextual patterning to their Spanish laterals. It remains to be seen if the results will show an English-like context-specific allophonic patterning in their Spanish (Barlow, 2014) or a Spanish-like lateral in both syllable onset and coda (Solon, 2016).

As bilingual language mode is considered to occur in situations that simultaneously activate the two languages, I assume that a bilingual mode can be induced experimentally if the stimuli come from both languages and the task asked of the participants requires processing of both languages. With respect to language mode, the main hypothesis is that the effects of tran-

The results in Grosjean and Miller (1994) suggest that the phonetic switch from one language to another is seamless and cost-free, but other laboratory studies have shown some type of language mode effect in phonetic production, ranging from moderate to significant (Antoniou, Best, Tyler, & Kroos, 2010, 2011; Bullock et al., 2006; Olson, 2012; Simonet, 2014). As a result, I expect to find F2–F1 values for both languages converging in bilingual mode for speakers that otherwise would maintain acoustically distinct laterals in each language when producing them in monolingual mode. This study is particularly well-positioned to shed light on how language dominance interacts with language mode effects. As such, one of the main contributions and novelty of the present study is that it investigates language dominance and language mode in bilingual speech, two variables that have not previously been addressed concurrently in phonetic studies on heritage speakers and L2 learners.

3.1. Method

3.1.1. Participants

Sixty participants were recruited to participate in the production experiment. The sample comprised 40 female early Spanish-English bilinguals and 20 female late L2 Spanish learners who were undergraduate students enrolled in upper-division Spanish classes at the University of California, Santa Cruz. All participants reported normal speech and hearing and normal or corrected to normal vision, and each participant received a stipend for their participation in the study.

The early Spanish-English bilinguals ($N = 40$) consisted of Spanish heritage speakers who had been raised and educated in a bilingual environment in the United States, having extensive exposure to both Spanish and English on a daily basis. Following the categorization of Silva-Corvalán (1994), the early Spanish-English bilingual group was further divided into three groups as a function of their immigrant generation: Generation 1.5 (G1.5), Generation 2 (G2), and Generation 3 (G3). The G1.5 group ($N = 10$) consists of foreign-born (i.e., Mexico) females who arrived in the United States between the age of 6 and 11. The G2 group ($N = 10$) are individuals who were born in the United States with either one parent or both parents born in Mexico. Finally, the G3 group ($N = 20$) are Spanish-English bilinguals who, as their parents, were born in the United States, while their grandparents were born in Mexico. All participants had been raised speaking Spanish to various degrees at home with their parents and were completing their education in the U.S. The L2 Spanish learners ($N = 20$) had been raised in a monolingual English household in the United States, spoke English as their native language, and learned Spanish at school. They reported not being native speakers or fluent in any other language. The L2 learners were Spanish studies majors, they had completed their entire education in the U.S., and they were enrolled in upper-division Spanish courses at the time of testing. In order to be enrolled in upper-division Spanish courses, students are required to achieve a certain score on a placement test and advance beyond the six- an upper-intermediate level of university-level language study. Table 1 provides the age, age of exposure, accent self-ratings, and typical daily use of Spanish and English for each speaker group.

All participants completed the Bilingual Language Profile (BLP) questionnaire (Birdsong et al., 2012). The BLP is an instrument for assessing language dominance through self-reports and it produces a continuous dominance score and a general bilingual profile taking into account multiple dimensions: age of acquisition of the L1 and L2, frequency and contexts of use, competence in different skills, and attitudes towards each language. All of these factors are organized in four modules (language history, language use, language proficiency, and language attitudes), which received equal weighting (see Gertken, Amengual, & Birdsong, 2014). The BLP was administered prior to beginning the experiments, and was provided either in Spanish or English, depending on participant preference. The classification of participants as Spanish-dominant or English-dominant was determined by their responses to the questionnaire, which generated a language particular score for each module, a global score for each language, and a global score of dominance. The point system was converted to a scale score with the Spanish score subtracted from the English score. Dominance scores ranged from -72 to 137: negative values indicate Spanish-dominance whereas positive values indicate English-dominant. Fig. 1 provides the distribution of the G1.5, G2, G3, and L2 Spanish speakers as a function of their language dominance scores according to the BLP.

3.1.2. Materials and procedure

The production of the target Spanish and English /l/ in pre-vocalic and post-vocalic position was elicited in a reading-aloud task. The materials consisted of 30 experimental items for each language, and controlled for three factors: (1) vowel context, (2) syllable position, and (3) stress.[1] Three vowels in Spanish (/a/, /e/, /i/), three vowels in English (/e/, /i/, /iː/), and two phonological contexts (syllable-initial, syllable-final) were selected. These vowel contexts were used in order to avoid the context of a back vowel, which is an environment that is known to favor velarization (and also vocalization) of the lateral variant (Recasens, 2004; Recasens & Espinosa, 2005; Simonet, 2010). For instance, Solon (2016) found that L2 Spanish learners produced more velarized laterals in Spanish when following a back vowel than when the lateral followed a non-back vowel. The materials are provided in Table 2.

Each target item was embedded in a carrier phrase *Yo puedo decir* TARGETWORD (Spanish) or *I can say* TARGETWORD (English). In contrast to the syllable-initial condition, which is embedded within a phrase, the syllable-final condition is aligned with the phrase-final (utterance final) position, an

[1] Of the 60 experimental items in English and Spanish, 27 are monosyllabic, 27 are disyllabic, and 6 are trisyllabic words. It must be noted that most of the Spanish words are disyllabic whereas the majority of English words are monosyllabic. Given that monosyllabic words are typically longer than polysyllabic words (i.e., polysyllabic shortening), this length difference could have an effect on the acoustic realization of these laterals. Previous

Table 1
Age, age of exposure, accent self-ratings, and typical daily use of both languages for each group. Group averages and standard deviations (parentheses) are shown.

	G1.5 (*N* = 10)	G2 (*N* = 10)	G3 (*N* = 20)	L2 (*N* = 20)
Age	21.2 (0.5)	20.6 (1.2)	20.2 (1.1)	21.7 (0.2)
Age of exposure[a]	*E* = 8.2 (2.1)	*E* = 2.2 (1.3)	*E* = 0.4 (0.3)	*E* = 0 (0)
	S = 0 (0)	*S* = 0.3 (0.2)	*S* = 3.1 (1.1)	*S* = 9.1 (4.4)
Self-reported accent[a,b]	*E* = 7.8 (0.9)	*E* = 8.5 (0.3)	*E* = 8.8 (1.1)	*E* = 9.5 (0.2)
	S = 9.2 (0.4)	*S* = 7.6 (1.1)	*S* = 7.2 (1.3)	*S* = 6.8 (1.5)
Daily use[c]	7.1 (1.6)	5.5 (2.1)	4.9 (1.9)	2.2 (1.2)

[a] E = English; S = Spanish.
[b] 1 = non-native; 10 = native.
[c] 1 = only English; 9 = only Spanish.

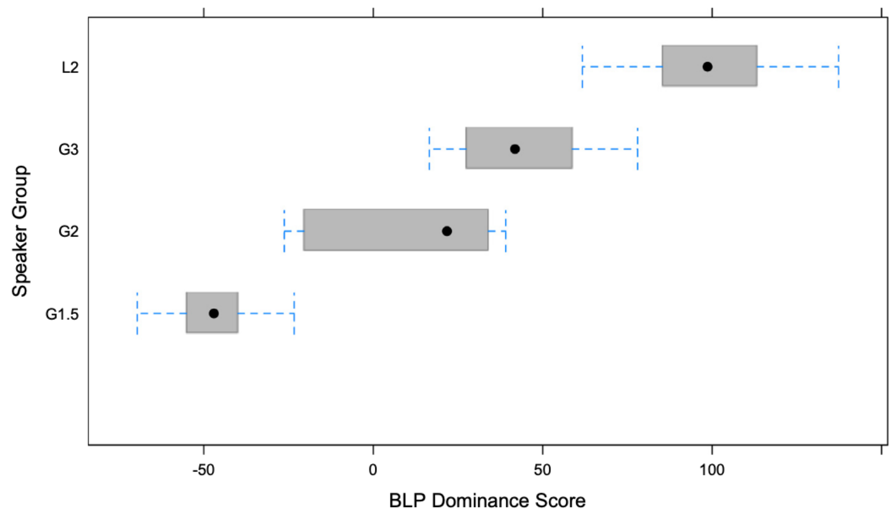

Fig. 1. Language dominance as a function of group according to the Bilingual Language Profile.

Table 2
Spanish and English stimuli.

Position/language	/a/	/e/	/i/
Syllable-initial (Spanish)	Labio 'lip'	Leche 'milk'	Libro 'book'
	Laca 'hairspray'	Lejos 'far'	Liso 'smooth'
	Lago 'lake'	Letra 'letter'	Libre 'free'
	Lazo 'bow'	Leve 'mild'	Liga 'league'
	Lata 'can'	Leña 'firewood'	Lima 'lime'
	/a/	/e/	/i/
Syllable-final (Spanish)	Mal 'bad'	Miel 'honey'	Civil 'civil'
	Capital 'capital'	Decibel 'decibel'	Raíl 'rail
	Cristal 'glass'	Burdel 'brothel'	Senil 'senile'
	Musical 'musical'	Nivel 'level'	Viril 'virile'
	Tribunal 'tribunal'	Papel 'paper'	Mil 'thousand'
	/ɛ/	/iː/	/ɪ/
Syllable-initial (English)	Lemon	Legal	Lick
	Led	Lethal	Liberty
	Lend	Leave	Literal
	Length	Leap	Lip
	Leg	Leach	Limo
	/ɛ/	/iː/	/ɪ/
Syllable-final (English)	Shell	Kneel	Ill
	Hell	Wheel	Grill
	Fell	Feel	Bill
	Bell	Veal	Kill
	Tell	Seal	Fill

used for other experiments. The crucial manipulation in the present study is that participants were asked to participate in three different experimental sessions. In the first two sessions, hereafter, the monolingual sessions, participants read words only in the target language: either Spanish or English. In the third session, the bilingual session, the Spanish and English carrier phrases were presented together in random order. That is, a given speaker might produce *Yo puedo decir* LIMA immediately followed by *I can say* LEMON. Within this recording session, participants produced equal numbers of Spanish and English experimental words. The main goal was to induce various levels of activation of each language of the bilingual individual in a monolingual and bilingual mode: Spanish monolingual session (30 words $\times$ 2 repetitions = 60), English monolingual session (30 words $\times$ 2 repetitions = 60), and Spanish/English bilingual session (60 words $\times$ 2 repetitions = 120). The same experimental words, which appeared in randomized order, were produced in the monolingual and bilingual sessions, and there was at least a 72-h interval between each experimental session.

As noted in Simonet (2014), it is important to acknowledge that it is likely that the English and Spanish monolingual sessions were unsuccessful in fully inhibiting the activation of the experimental sessions were conducted in the same setting and the same speakers participated in each of the three sessions with a Spanish-English bilingual research assistant. During the monolingual sessions, care was taken so that instructions were given exclusively in the language in which they were tested (i.e., Spanish or English) by two different research assistants: one that only gave instructions in English and the other who only gave instructions in Spanish. English was used to greet and provide instructions to the participants in the bilingual session by a third research assistant. It is very possible that the three experimental sessions induce speakers to some degree in a bilingual mode, but the assumption is that one of the three sessions is “more” bilingual than the other two.

The production task was conducted individually in a sound-attenuated booth with participants comfortably seated in front of a computer display. Participants were told that the study involved reading sentences on a computer screen and that their speech would be recorded for subsequent acoustic analysis. Each sentence was presented on a computer screen and participants were asked to read the sentences clearly and with a natural pace, speaking neither too quickly nor too slowly. In other words, participants were asked to produce “clearly enunciated speech” (see Kessinger & Blumstein, 1997, p. 147) in the production of a carrier phrase, thus, articulating at a slow rate. Even though the phonological context in which the /l/-initial words were elicited was not identical between Spanish and English, it is important to note that the participants were instructed to introduce a pause at the word boundary, which minimized the possible effects of phonological context in the production of these word-initial laterals. The speech samples were recorded using a head-mounted microphone (Shure SM10A) and an audio interface (MOTU Ultra Like mk3), digitized (44 kHz, 16-bit quantization), and computer-edited for subsequent acoustic analysis.

3.1.3. Acoustic analysis

The acoustic analysis was based on the investigation of the F1 and F2 values extracted at the midpoint of the lateral consonants. F2 is considered the primary acoustic correlate of degree of velarization in alveolar laterals (Carter & Local, 2007; Recasens & Espinosa, 2005). The Spanish and English laterals were segmented in Praat (Boersma & Weenink, 2016) using synchronized waveforms and spectrographic displays. Praat scripts were run to parse each participant’s recording into individual files for each target item, and text grids were created by manually marking the lateral in each token. Formant trajectories, especially the trajectory of the second formant (F2), as well as intensity displays, were taken as indicators of lateral onsets and offsets. The lateral onset was marked at the first pitch period in which F2 showed a noticeable decrease in intensity than that corresponding to the preceding vowel in syllable-final laterals, usually coupled with a drop in the amplitude of the waveform (White & Mattys, 2007), or at the first the last pitch period in which F2 could be observed when in syllable-final position.[2]

Formant tracks were calculated with the Burg algorithm as built into the Praat program. The effective window length for the calculation was set at 25 ms, and was maintained across tokens and speakers. The maximum number of formants to be located by the formant tracker was always five, and the ceiling was set at 5.5 kHz, in order to minimize tracking errors (Escudero, Boersma, Rauber, & Bion, 2009). These parameters were selected after careful spectrographic inspection. Formant values were extracted in Hertz and were further converted to Bark, using the Hz-to-Bark function available in Praat. The bark scale is a logarithmic psychoacoustic scale that ranges from 1 to 24, and is a measure of frequency based on the critical bandwidths of hearing believed to reflect human perception (Johnson, 2003; Traunmüller, 1990; Zwicker, 1961).

The analysis of the Spanish and English laterals in this study consist of the F2–F1 differences, in Bark. This measure incorporates F1 values into the parameter while maintaining F2 values. Darker (more velarized) laterals are expected to have lower F2 values, higher F1 values, and thus smaller F2–F1 distances than clear laterals (Recasens & Espinosa, 2005). The effects of vocal tract-size differences caused by sex on the acoustics of laterals were minimized because the participant sample consisted exclusively of female speakers. This reduces the need for inter-speaker acoustic normalization procedures (Adank, Smits, & van Hout, 2004).

3.1.4. Dataset

The three sessions yielded a total of 240 /l/ productions per participant, for a total of 14,400 measurements. However, 5.4% of the data (N = 789) had to be excluded due to recording errors, mispronunciations, or because laterals were produced in a way that interfered with the reliable identification of the onset and/or offset of the lateral variant (i.e., there was no clear break between the vowel and the lateral). As a result, the dataset comprised a total of 13,611 measurements. In order to examine the effects of language, language dominance, and language mode in the acoustic realization of these early and late Spanish-English bilinguals, datasets of by-subjects aggregates were created including the median F2–F1 distances over subjects as a condition of language, syllable position, and language mode (six values per subject, two per language per language mode condition and syllable position). Taking the median rather than the mean has been claimed to reduce the effect of formant measurement errors (Escudero et al., 2009).

[2] It should be noted that the immediately preceding sound in the Spanish carrier phrase is the tapped/flapped /r/ in the word *decir* ‘to say’, which could cause the appearance of an epenthetic vowel between the tap and the following consonant making it difficult to distinguish the end of the epenthetic vowel from the beginning of the /l/. Because participants were instructed to make a pause at the word boundary, the possible effects of coarticulation in the pronunciation of these word-initial laterals were greatly reduced.

3.2. Results

The complete dataset containing F2–F1 values was submitted to a mixed-model ANOVA, which was performed using R (R Core Team, 2017), with speaker group (G1.5, G2, G3, and L2 Spanish learners) as between-subjects factor, language (Spanish, English), syllabic position (word-initial, word-final) and language mode (monolingual, bilingual) as within-subjects factors, and subject and item as the random terms. The mixed-design ANOVA yielded significant main effects of *speaker group* ($F(3,56) = 24.58$, $p < 0.001$), *language* ($F(1,392) = 868.02$, $p < 0.001$), and *syllabic position* ($F(1,392) = 220.24$, $p < 0.001$), but *language mode* on its own yielded no significant main effect ($F(1,392) = 3.31$, n.s). In addition, there were significant interactions between speaker group and syllabic position ($F(3,392) = 2.94$, $p < 0.05$); speaker group and language ($F(3,392) = 4.27$, $p < 0.01$); syllabic position and language ($F(1,392) = 384.93$, $p < 0.001$); speaker group and language mode ($F(3,392) = 20.47$, $p < 0.001$); language and language mode ($F(1,392) = 110.60$, $p < 0.001$); and a three-way interaction between speaker group, language, and language mode ($F(3,392) = 9.54$, $p < 0.001$). The results of each fixed factor and the interactions of these factors motivated the division of the dataset into smaller subsets in order to allow for the investigation of the specific research questions. For this reason, the results of the mixed-effects model and the interactions explored with pairwise comparisons are reported separately below to enable the discussion of each research question addressed in this study. In all cases, paired *t*-tests and effect sizes calculated by means of paired Cohen's *d* are reported.

3.2.1. Language

The first research question asked whether Spanish heritage speakers and L2 learners produce acoustically similar Spanish and English lateral approximants or if they maintain separate acoustic realizations for each language. As can be observed in Fig. 2, each group maintains a language-specific phonetic distribution in their acoustic realization of the lateral approximants. Pairwise comparisons revealed significant differences in the acoustic realization of Spanish and English laterals produced by G1.5 speakers (diff. = −2.54, $t(39) = -5.96$, $p < 0.001$, Cohen's $d = 0.92$), G2 speakers (diff. = −2.33, $t(39) = -6.76$, $p < 0.001$, Cohen's $d = 1.07$), G3 speakers (diff. = −2.42, $t(79) = -10.84$, $p < 0.001$, Cohen's $d = 1.21$), and L2 Spanish learners (diff. = −1.86, $t(79) = -8.66$, $p < 0.001$, Cohen's $d = 0.96$).

The interaction between speaker group and language was further explored by analyzing the effects of group separately for each language. Bonferroni-corrected, paired *t*-tests compared the acoustic realization of each group in English and Spanish. The alpha level was adjusted accordingly (0.05/6 = 0.0083). In English, these pairwise comparisons yielded significant differences between L2 speakers and G3 speakers (diff. = 0.68, $t(79) = 4.62$, $p < 0.001$, Cohen's $d = 0.51$); L2 speakers and G2 speakers (diff. = 1.17, $t(79) = 4.62$, $p < 0.001$, Cohen's G1.5 and G2 speakers (diff. = 0.47, $t(39) = 1.83$, n.s); G2 and G3 speakers (diff. = 0.48, $t(80) = 1.46$, n.s); and G1.5 and G3 speakers (diff. = 0.95, $t(59) = 2.2$, n.s).

The Spanish data were also explored in a series of Bonferroni-corrected pairwise comparisons, and the alpha level was also adjusted to 0.0083. These comparisons yielded significant differences between G1.5 speakers and G2 speakers (diff. = 0.68, $t(39) = 3.62$, $p < 0.001$, Cohen's $d = 0.57$); G3 speakers and L2 speakers (diff. = 1.24, $t(79) = 7.05$, $p < 0.001$, Cohen's $d = 0.78$); G1.5 and G3 speakers (diff. = 1.06, $t(79) = 5.89$, $p < 0.001$, Cohen's $d = 0.82$); G1.5 and L2 speakers (diff. = 2.31, $t(112) = 10.41$, $p < 0.001$. Cohen's $d = 1.26$); and G2 and L2 speakers (diff. = 1.63, $t(109) = 7.18$, $p < 0.001$, Cohen's $d = 1.01$), but there was no significant difference between G2 and G3 speakers (diff. = 0.38, $t(75) = 2.04$, n.s).

To summarize, Spanish and English laterals, as produced by G1.5, G2, G3, and L2 Spanish speakers, differ in their degree of velarization. Specifically, all groups produce darker or more velarized (smaller F2–F1 distances) English laterals and clearer or less velarized (greater F2–F1 distances) Spanish laterals. Additionally, all speaker groups differ from each other with respect to the pronunciation of the Spanish laterals, except for G2 and G3, and the L2 Spanish group differs from the other three heritage speaker groups in the acoustic realization of the English laterals. These results point to an effect of language dominance, which seems to be more sensitive to the articulation of the Spanish laterals, as each group from more Spanish-dominant (G1.5) to more English-dominant (L2) gradually produces a darker Spanish lateral. In English, there is only a significant difference between the most English-dominant group (L2) and the early bilingual groups, the Spanish heritage groups (G1.5, G2, and G3).

3.2.2. Syllable position

The second research question inquired about the effect of syllable position in the acoustic realization of Spanish and English laterals. Specifically, the goal was to examine if these bilingual groups acquired the English-specific contextual (allophonic) patterns of use of these laterals. And if so, if they were transferred to their Spanish productions. As a reminder, in English a dark /l/ (i.e., [ɫ]) is expected in coda positions, whereas a light /l/ (i.e., [l]) is expected to occur in the syllable onset. In other words, laterals are typically darker in syllable rhyme and clearer in syllable onset. In contrast, the Spanish /l/, which is generally clear, does not vary by syllabic context and is assumed to be relatively consistent. Fig. 3 presents the F2–F1 (Bark) values in the productions of lateral variants in word-initial (WI) and word-final (WF) position as a function of group and language.

The statistical model revealed a significant main effect of syllabic position, and significant interactions between syllabic position and language, and syllabic position and speaker group. In order to explore these interactions, a series of four two-sample, Bonferroni-corrected, paired *t*-tests compared the acoustic realization of each speaker group in word-initial and word-final position in Spanish and English separately.

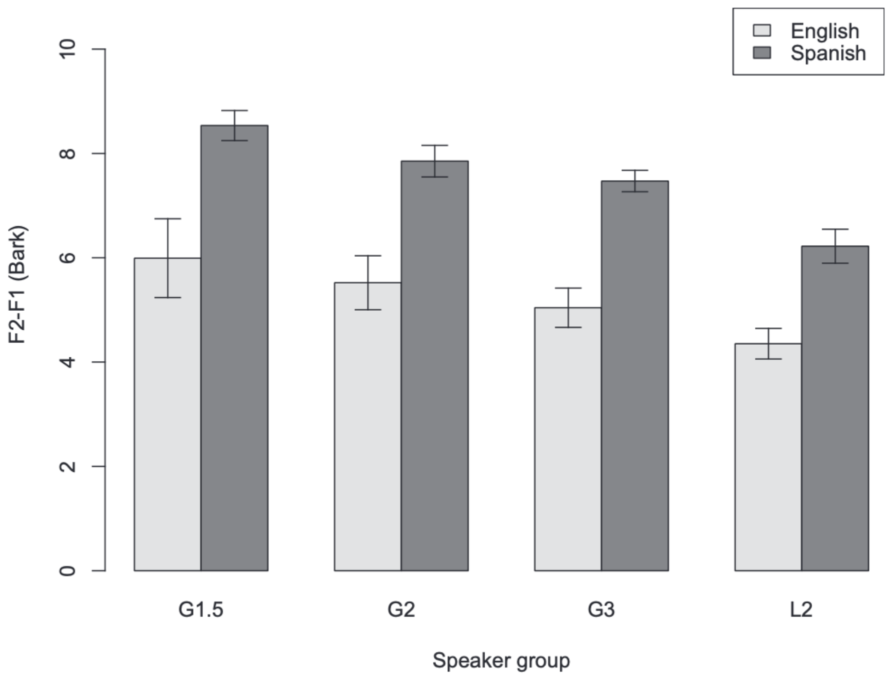

Fig. 2. F2–F1 (Bark) values as a function of group and language.

speakers (diff. = 3.75, $t(19) = 11.83$, $p < 0.001$, Cohen's d = 2.64), G2 speakers (diff. = 2.37, $t(19) = 8.62$, $p < 0.001$, Cohen's $d = 1.92$), G3 speakers (diff. = 2.78, $t(39) = 13.16$, $p < 0.001$, Cohen's $d = 2.09$), and L2 speakers (diff. = 2.02, $t(39) = 10.22$, $p < 0.001$, Cohen's $d = 1.61$). Not surprisingly, these results show that all speaker groups, who are native American English speakers, have acquired the contextual patterning of English whereby lighter /l/s are produced in syllable onset position and darker /l/s are produced in syllable coda. The Spanish data in word-initial and word-final position were also further investigated in Bonferroni-corrected pairwise comparisons. These comparisons also yielded significant differences in the acoustic realization of laterals as a function of syllable position produced by G1.5 speakers (diff. = −0.61, $t(19) = -6.41$, $p < 0.001$, Cohen's $d = 1.41$), G2 speakers (diff. = −0.69, $t(19) = -4.1$, $p < 0.001$, Cohen's $d = 0.90$), and G3 speakers (diff. = −0.44, $t(39) = -3.3$, $p < 0.01$, Cohen's $d = 0.51$). However, there was not a significant difference in the production of word-initial and word-final laterals by L2 speakers (diff. = 0.01, $t(39) = 0.07$, n.s). Even though the results of these post hoc analyses reveal effects of syllable position in the production of Spanish /l/, it is important to note that in addition to not being as robust as the contextual distribution in English, it is in the opposite direction of what would be expected in English: the Spanish heritage speakers produced lighter laterals in word-final position, instead of a darker realization. Therefore, it is safe to assume this is not a case of transfer of the English lateral allophonic patterning from English to Spanish. Finally, as the F2–F1 values differed as a function phrase final (utterance-final) position, as this confound was present in both English and Spanish materials alike.

3.2.3. Language mode

The third research question asked about the effects of language mode in the acoustic realization of English and Spanish laterals. Specifically, the experiment was designed to explore if there was evidence of phonetic convergence of the laterals in each language as a result of eliciting a bilingual language mode. Fig. 4 presents the F2–F1 (Bark) values in the productions of the lateral variants in monolingual mode and bilingual mode as a function of group and language. Language mode was a significant factor affecting the F2–F1 values in the statistical model. The significant interaction between language mode and speaker group, between language mode and language, and between language mode, speaker group, and language in the statistical model shows that language mode affected each speaker group and language differently.

The effects of language mode for each language is again analyzed separately for each speaker group. Bonferroni-corrected, paired t-tests compared the acoustic realization of the laterals produced by each group in English and Spanish. The alpha level was adjusted (0.05/8 = 0.0062). The pairwise comparisons investigating the effects of language mode in the productions of Spanish laterals by the L2 Spanish speaker group yielded a significant difference (diff. = −1.45, $t(39) = -18.02$, $p < 0.001$, Cohen's $d = 2.84$), but this language mode difference was not found in their English lateral production (diff. = 0.07, $t(39) = 1.2$, n.s). The analysis of the G3 speaker

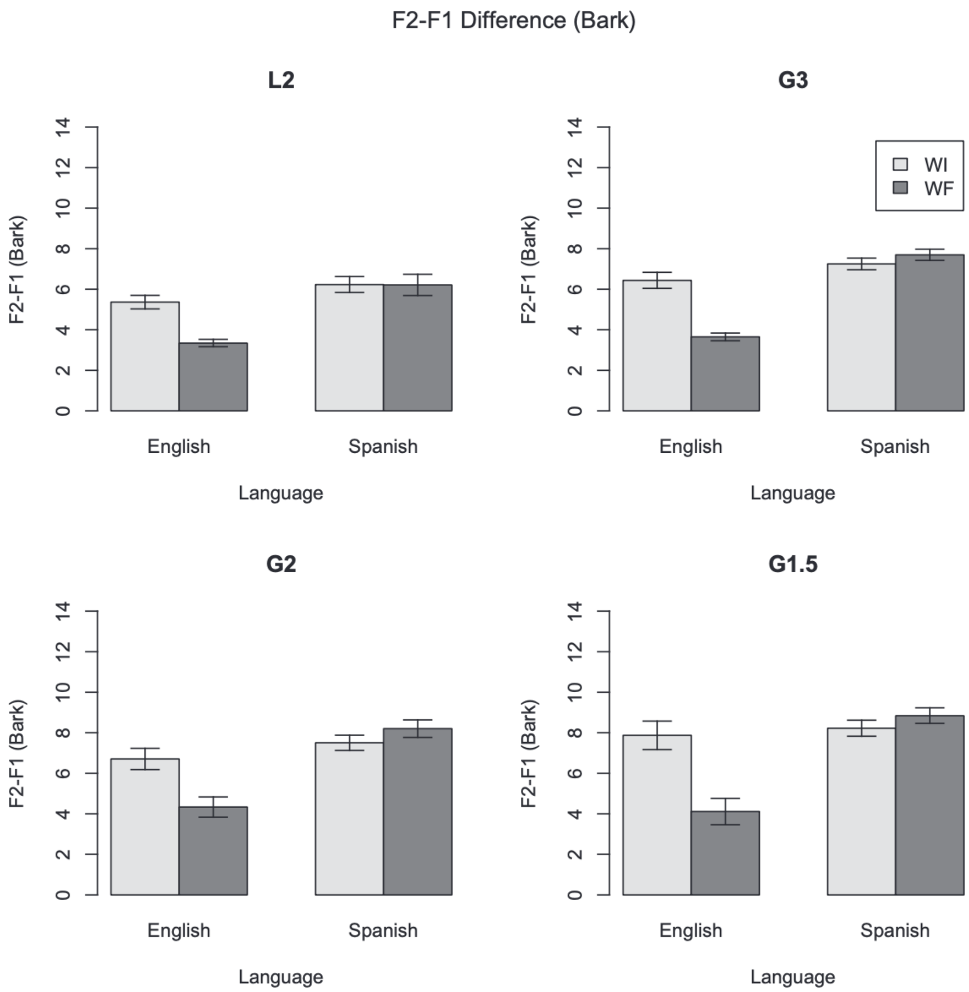

Fig. 3. F2–F1 (Bark) values of productions in word-initial (WI) and word-final (WF) position as a function of group and language.

and in English (diff. = 0.38, $t(39) = 6$, $p < 0.001$, Cohen's $d = 0.95$). A similar significant effect between the production of laterals in monolingual mode and bilingual mode was also found in the Spanish (diff. = −0.78, $t(19) = -5.09$, $p < 0.001$, Cohen's $d = 1.14$) and English (diff. = 0.92, $t(19) = 3.8$, $p < 0.01$, Cohen's $d = 0.85$) productions of the G2 speaker group. Finally, for the G1.5 speaker group, there was a significant difference between the English laterals produced in monolingual mode and bilingual mode (diff. = 2.13, $t(19) = 19.2$, $p < 0.001$, Cohen's $d = 4.34$), but this language mode difference was not found in their Spanish (diff. = −0.03, $t(19) = -0.46$, n.s).

These results show that the effects of language mode are robust, but that they affect each speaker group differently. Specifically, the G2 and G3 group, which according to their BLP dominance scores are slightly English-dominant, showed an effect of language mode in both of their languages. The data from the G1.5 speaker group, the only Spanish-dominant group in the sample, indicated that language mode dominance score according to the BLP, produced acoustically indistinct English laterals in monolingual mode and bilingual mode; however, they were found to produce more velarized (i.e., English-like) laterals in Spanish when in bilingual mode compared to monolingual mode. As with the pattern found in the production of the G1.5 group, bilingual mode was found to only have an effect on the acoustic realization of laterals in the weaker language.

4. Discussion

The present study analyzed the acoustic realization of the Spanish and English voiced lateral consonant /l/ in the production of Spanish-English bilinguals, classified as a function of their linguistic background: early and late Spanish-English bilinguals. The early Spanish-English bilinguals were Spanish heritage speakers who had been raised and educated in a bilingual environment in the United States, having extensive exposure to both Spanish and English on a daily basis, and

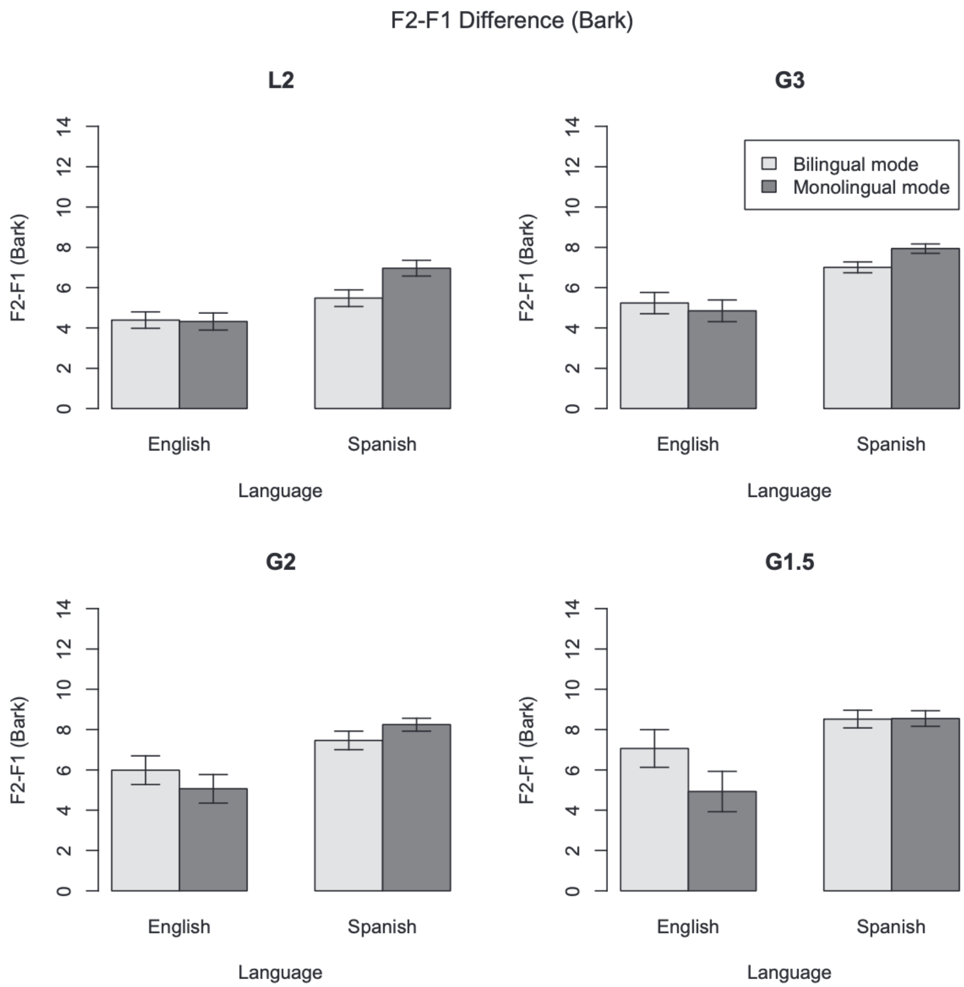

Fig. 4. F2–F1 (Bark) values of productions in monolingual mode and bilingual mode as a function of group and language.

further divided into three groups as a function of their immigrant generation: Generation 1.5, Generation 2, and Generation 3. The present study explored the effects of syllable position, language dominance, and language mode in the acoustics of the laterals produced by these early and late Spanish-English bilinguals.

The results from the production task indicate that there is a language-specific phonetic distribution in the acoustic realization of Spanish heritage speakers and L2 Spanish learners: Spanish and English laterals, as produced by each speaker group, differed in their degree of velarization. The data show a darker or more velarized articulation (lower F2 and smaller F2–F1 distances) in English laterals, and clearer or less velarized production (higher F2 and greater F2–F1 distances) in Spanish laterals for each of the speaker groups included in this study. This demonstrates that all participants had acquired the distinct acoustic properties of the laterals in Spanish and English. The fact that the early and late bilinguals in this study were able to establish new L2 categories for their laterals may be of new category formation. It is important to note, however, that there were differences in the degree of velarization between the groups in that dominant and non-dominant speakers of each language differed in the "lightness" or "darkness" of these laterals. Particularly for the Spanish heritage speaker group, the laterals were produced lighter (i.e., more Spanish-like) by G1.5 speakers, and darker (i.e., more English-like) by G3 speakers. The fact that laterals in both languages became more velarized (i.e., darker) as a function of being more dominant in English is in line with the predictions that arise from the well-documented process of language shift from Spanish to English in the United States within three generations (Bills et al., 1995; Martínez, 2009; Potowski, 2004). These language dominance effects add to previous findings that show a gradient interlingual influence as a result of age of acquisition (Barlow, 2014) and language proficiency (Solon, 2016).

A large majority of previous studies have been concerned with the acquisition of language-specific phonemic contrasts; however, less is known about the bilingual acquisition of nonthese variants in each language. The analysis of Spanish and English laterals enables the examination not only of the acquisition of fine-grained phonetic detail, but also the application of the allophonic alternation in only one of their languages (i.e., English). Do these Spanish-English bilinguals acquire the English-specific contextual (allophonic) patterns of use of the lateral variants and is this allophonic patterning in English transferred to their Spanish production? The acoustic data showed that all groups, independently of their language dominance, produced a more velarized English lateral in syllabic coda (word-final position) than in syllabic onset (word-initial position).[3] Importantly, and contrary to the findings in Barlow (2014) and Solon (2016), this context-specific allophonic patterning in English was not transferred to their Spanish. This is evidence that these early and late bilinguals have formed separate phonetic categories for Spanish and English /l/s in each context. These bilinguals acquired the phonetic details of laterals in Spanish and English with respect to lateral darkness (as measured by F2–F1) and it was accompanied by the development of language-specific allophonic patterns. In other words, these bilinguals presented a bimodal representation of English /l/ with realizations of this phone being light and dark as a function of syllable position, and a monomodal representation of Spanish /l/, being a lighter articulation than in English, regardless of syllable position. Therefore, all of these Spanish-English bilinguals produce target-like language-specific phonetic detail in their laterals but also acquire patterns and categories that demonstrate phonological learning. This may not be a surprising finding for highly proficient bilinguals if we consider that the lateral distribution in English is a predictable occurrence as it occurs in a context of complementary distribution. It is reasonable to expect that phones, such as the English laterals, with a predictable distribution, attract attention, which in turn facilitates phonetic and phonological learning.

The final research question that this experiment was designed to explore was the following: did the activation of both phonological systems, as induced by the experimental setting, have an effect on the production of the alveolar laterals in Spanish and English? The analyses of the data produced in the monolingual and bilingual sessions indicated that the lateral production of these bilinguals differed as a function of the experimental session or the speech production setting in which the words were uttered. Resorting again to the predictions of the SLM, even though the results show that these bilinguals maintained two separate distributions for the lateral variants in their dominant and non-dominant language, presenting evidence of new category formation, it must be noted that participants transferred phonetic characteristics of the laterals produced in their dominant to their non-dominant language, when articulating these laterals in bilingual mode. It can be argued that this would present evidence of category assimilation (Flege, 1987). In this case, the SLM predicts that when the L2 phonetic category is sufficiently similar to an L1 phone, it results in equivalence classification. This category assimilation gives rise to a merged L1 and L2 phonetic category, which differs from that of a monolingual speaker of either language. This assimilation is predicted to result in a difficult discrimination and accented production of these merged speech sounds.

Importantly, the data display asymmetrical effects as a function of language dominance: when in bilingual mode, all four groups of Spanish-English bilinguals produced a less target-like lateral in their non-dominant language. In other words, the acoustic realization of the lateral in the non-dominant language shifted (in terms of the F2–F1 values) in the direction of the lateral in the dominant language. According to the Language Mode framework (Grosjean, 1982, 1989, 1998, 2001, 2008), the linguistic behavior of bilinguals is affected by the communicative context. The results of the present study suggest that interlingual phonetic interactions are indeed affected by the relative degree of language activation, resulting in a "gestural drift" (Sancier & Fowler, 1997), even in relatively short-term changes in the linguistic environment. A bilingual individual, especially one with high proficiency in both languages, should have the ability to alternate between a light lateral in Spanish and a light and dark lateral when speaking in English. However, the results of this study confirm that there is a cost in bilingual language processing: activating the words of one language has immediate effects on the phonetics of the words of the other language (Antoniou et al., 2010; Elman, Diehl, & Buchwald, 1977; Flege & Eefting, 1987; García-Sierra, Diehl, & Champlin, 2009; Hazan & Boulakia, 1993).

The current theoretical speech learning models of cross-linguistic perception and production (Perceptual Assimilation Model, PAM; Perceptual Assimilation Model of Second Language Speech Learning, PAM-L2; the Speech Learning Model, SLM; and the Second Language Perception Model, L2LP) make straightforward predictions about the learnability of L2 speech sounds depending on the perceived similarity between the sounds of the L1 and L2. However, these models do not address how speech perception and production may be affected by the situational language context in bilinguals. Therefore, the models available do not offer predictions as to which factors facilitate a shift in the characteristics of phonetic categories as a result of language activation. In other words, there is a lack of a theory of speech perception and production that explicitly considers the contribution of the situational language context in bilinguals. Such effects must be operationalized in a model of bilingual speech that accounts for the variable production patterns found as a result of language dominance and language mode.

A model of bilingual phonological representation and processing based on the principles of episodic (or complementary-systems) frameworks could explain these findings (Goldinger, 1997, 1998). Assuming that lexicons in different languages are mentally interconnected (Costa, Santesteban, & Caño, 2005; Jarvis & Pavlenko, 2008), lexical

[3] It must be acknowledged that due to being produced in utterance-final position, the acoustic realization of laterals in syllable-final position may have been influenced by preboundary lengthening effects. In contrast, the target items in syllable-initial position are likely to be embedded within a phrase, so they would not be subject to the boundary-edge

Pierrehumbert, 2001; Pierrehumbert, 2003a, 2003b) are theoretical frameworks that are able to account for the effects of the increased activation of both languages as a result of being in bilingual mode. An expanded model that includes bilingual data would assume that words, or exemplars, in a bilingual lexicon develop connections across languages, and lexical processing activates the words of both languages non-selectively. An increase in the strength of activation of words in bilingual mode could be responsible for displacing by means of assimilation, and resulting in a "gestural drift", the degree of velarization, of interconnected clouds of data.

Exemplar models assume that speech perception and production are closely linked, with perceived instances of a word or speech sound being the basis for production targets. Clusters of similar experiences—that is, of "exemplars"—are formed including productions that share a particular acoustic property. These exemplars are categorized by their similarity to extant stored exemplars so that clouds of memory traces group similar exemplars close to each other while dissimilar ones are more distant. The exemplars themselves include much more than just purely phonetic information: the representation of that word includes its meaning(s) and all the acoustic, lexical, social, and contextual information from the perceptual event (Ettlinger & Johnson, 2010). Such a model can account for how speakers might possess fine-grained, detailed, and word-specific knowledge about the sounds and words of their language and require no phonological abstraction prior to lexical access (Coleman, 2002; Johnson, 2007; Pierrehumbert, 2001). The exemplar model explains production as drawing on these clouds of perceptual events to choose an appropriate phonetic target that may vary based on the information stored with the representation, such as social and contextual (i.e., language mode) demands.

The application of an exemplar-based approach to the production of these highly proficient Spanish-English bilinguals would assume mostly distinct exemplar clouds representing English and Spanish. This is reflected in the acoustic realization of laterals in monolingual mode. However, since these clouds are organized by the phonetic similarity of the exemplars and also include contextual information, there is likely to be overlap between the two otherwise independent language systems with respect to the articulation of words in bilingual mode. Thus, production of words in bilingual mode potentially draws from both English and Spanish exemplars instead of restricting the possible targets to the language-specific exemplars available for each language separately.

The formation of exemplars is undoubtedly affected by the quantity and quality of the input that the bilingual individual is exposed to in the environment, which explains the variable exemplars that bilinguals form depending on their linguistic experience. The phonetic detail to which a speaker/hearer is exposed during their lifetime is especially relevant in the acquisition of L2 phonology, as noted in Best (2011, p. 13): "native adult perceivers as well as L1-learning children and adult L2 learners exploit, rather than avoiding or being overwhelmed by, the multiple dimensions of natural phonetic variation in long-term increase in L2 production intelligibility (Bradlow, Akahane-Yamada, Pisoni, & Tohkura, 1999; Bradlow, Pisoni, Akahane-Yamada, & Tohkura, 1997). These studies reveal a benefit for L2 learners as well as highly experienced, fluent L2 speakers (Iverson, Pinet, & Evans, 2012), such as the highly proficient Spanish-English bilinguals included in this study.

A limitation of the present study is that it does not control for the effects of cognate status in Spanish and English, where we might expect more cross-linguistic influence in each language when producing and perceiving cognates versus non-cognates (Amengual, 2012, 2016b; Brown & Amengual, 2015; Tessel, Levy, Gitterman, & Shafer, 2018). In addition to carefully controlling cognate status, future studies should examine the production of laterals in monosyllabic, disyllabic, and trisyllabic words, where we expect to find polysyllabic shortening. Future work will also benefit from incorporating a closer examination of the phonetics-prosody interface, particularly in regard to boundary-related prosodic strengthening (Cho, 2004; Cho, Kim, & Kim, 2015; Cho & McQueen, 2005; de Jong, 1995), such as preboundary lengthening or domain-initial strengthening in the acoustic realization of laterals in bilingual speech. This will provide a relevant contribution to the fields of L2 phonetics and phonetic convergence, as previous results imply that "the detailed linguistic function of prosodic strengthening and its enhancement pattern of linguistic contrast is language specifically determined by making reference to other components of the linguistic system of the language such as the prominence system, the allophonic rules, and the phonetic/phonological feature system" (Cho, 2016, p. 136).

Another important limitation is that this study is not able to evaluate Spanish and English regional dialect features, which would contribute to the formation of the Spanish and English lateral phonological categories. Because Chicano English is characterized as having a clearer /l/ than other dialects of English (Frazer, 1996; Van Hofwegen, 2009), it is difficult to tease apart if speakers are producing a lighter /l/ because of the bilingual's knowledge of Spanish or if it is the influence of the Chicano English dialect. Without being able to account for these possible sociolinguistic effects, the results of this study show that these highly proficient bilinguals maintain language-specific phonological categories when producing laterals in Spanish and English. But also that interlingual phonetic interactions take place during bilingual speech processing, causing transient interference in the production of laterals as a result of the activation of both languages of the bilingual individual. Future studies would also benefit from including Spanish and English monolinguals to provide a baseline in the degree of velarization expected in English and Spanish for those individuals who do not have to inhibit the influence of their other language. However, care will have to be taken so that the particular participants are part of the same region (and exposed to the same varieties of Spanish and English) as the bilinguals in the study.

5. Conclusions

immigrant generation (i.e., G1.5, G2, G3), and late L2 Spanish learners. This experiment investigated the effects of variables not previously addressed concurrently on heritage speakers and L2 Spanish speakers, namely language dominance and language mode. Additionally, the present study contributes to the literature on L2 phonological acquisition and bilingualism by examining the acquisition of non-contrastive speech sounds or allophones in Spanish and English, which for these bilingual speakers, involves the inhibition of an allophonic distinction in English, as opposed to the acquisition of a new phonemic category. The results of this experiment indicate that these four groups of Spanish-English bilinguals that differ in their linguistic experience (i) maintain distinct laterals in their productions of each language, (ii) acquire both the acoustic properties of the particular lateral variants in the dominant and non-dominant languages as well as the specific contextual patterns of use of these variants in each language, and (iii) produce a less target-like lateral when in bilingual mode, in that the acoustic realization of the lateral in the non-dominant language shifts in the direction of the lateral variant in the dominant language. It has been proposed that a model of bilingual processing based on the principles of episodic frameworks, in which exemplars in a bilingual lexicon develop connections across languages and lexical processing activates the words of both languages non-selectively, can explain these findings.

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